Table 7.1 Differences between storing data with S3 and Glacier

	S3 Standard	S3 Glacier Instant Retrieval	S3 Glacier Flexible Retrieval	S3 Glacier Deep Archive
Storage costs for 1 GB per month in US East (N. Virginia)	$0.023	$0.004	$0.0036	$0.00099
Costs for 1,000 write requests	$0.005	$0.02	$0.03	$0.05
Costs for retrieving data	Low	High	High	Very High
Accessibility	Milliseconds	Milliseconds	1–5 minutes/3–5 hours/5–12 hours	12 hours/48 hours
Durability objective	99.999999999%	99.999999999%	99.999999999%	99.999999999%
Availability objective	99.99%	99.9%	99.99%	99.99%

The potential savings for storage costs are enormous. So what's the catch?

First, accessing data stored on S3 by using the storage classes S3 Glacier Instant Retrieval, S3 Glacier Flexible Retrieval, and S3 Glacier Deep Archive is expensive. Let's assume, you are storing 1 TB of data on S3 and decided to use storage type S3 Glacier Deep Archive. It will cost you about $120 to restore 1 TB of data stored in 1,000 files.

Second, fetching data from S3 Glacier Flexible Retrieval and S3 Glacier Deep Archive takes something between 1 minute and 48 hours, depending on the storage class and retrieval option.

Using the following example, we would like to explain what it means not to be able to access archived data immediately. Let's say you want to archive a document for five years. You do not expect to access the document more than five times during this period.